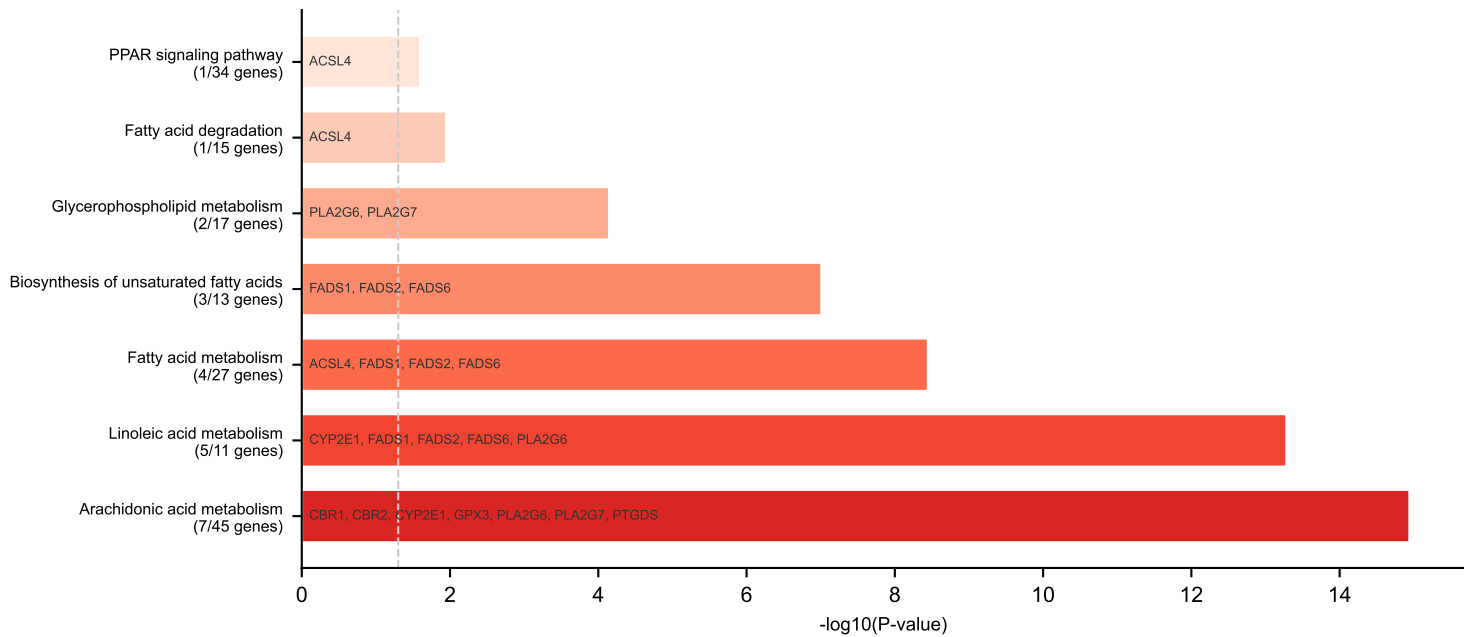


F. KEGG Pathway Enrichment (A ∩ B Genes)



FINAL CANDIDATE GENES — AA Liver-Muscle Axis

ACSL4 [AA□□/□□□□Lands cycle] Pattern: DLY75□□□TFB45□□□	↔ TRIM63	r=0.746	P=0.0337
CBR2 [COX→□□□□/□□□] Pattern: TFB45□DLY75□□□□	↔ TRIM63	r=0.927	P=0.0009
CYP2E1 [CYP450→EET/HETE] Pattern: DLY75□□□TFB45□□□	↔ FBXO32	r=0.852	P=0.0072
CYP2U1 [CYP450→EET/HETE] Pattern: DLY75□□□TFB45□□□	↔ TRIM63	r=0.772	P=0.0247
CYP4V2 [CYP450→EET/HETE] Pattern: DLY75□□□TFB45□□□	↔ FBXO32	r=0.772	P=0.0247
FADS1 [□□□□□LA→AA] Pattern: TFB45□DLY75□□□□	↔ TRIM63	r=0.753	P=0.0312
FADS2 [□□□□□LA→AA] Pattern: DLY75□□□TFB45□□□	↔ TRIM63	r=0.795	P=0.0183
FADS6 [□□□□□LA→AA] Pattern: TFB45□DLY75□□□□	↔ TRIM63	r=0.843	P=0.0086
GPX3 [LOX→□□□/HETE] Pattern: TFB45□DLY75□□□□	↔ TRIM63	r=0.839	P=0.0092
PLA2G6 [□□□□□AA] Pattern: DLY75□□□TFB45□□□	↔ TRIM63	r=0.808	P=0.0152
PTGDS [COX→□□□□/□□□] Pattern: TFB45□DLY75□□□□	↔ TRIM63	r=0.713	P=0.0472

Screening Pipeline: 81 AA genes → 51 PD-aligned → 15 sig cross-tissue → 11 final

Top KEGG Pathways: Linoleic acid metabolism. Arachidonic acid metabolism, Fatty acid metabolism